INCH-POUND
MIL-DTL-13253E
10 January 2007
SUPERSEDING
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26 June 1974

DETAIL SPECIFICATION

HANDSETS, GENERAL SPECIFICATION FOR

Inactive for new design after November 2, 1998

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for handsets (see 3.1, 6.1, 6.4).

1.2 Classification. Handset is of the following type, as specified (see 6.2).

1.2.1 Type. The type of handset is as follows:

MIL-DTL-13253/4	-	Handset, Type H-23( )/U

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-P-642	-	Plugs, Telephone, and Accessory Screws; General Specification for.
MIL-M-13189	-	Microphone Elements, Carbon and Earphone Elements, Magnetic (For Telephone and Radio Use).
MIL-F-14072	-	Finishes for Ground Based Electronic Equipment.

Comments, suggestions, or questions on this document should be addressed to: Defense Supply Center, Columbus, Attn: VAI, P.O. Box 3990, Columbus, Ohio, 43218-3990 or emailed to sound@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-202	-	Test Methods for Electronic and Electrical Components Parts.
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY (CAGE 80063)

SC-D-1041	-	Cord CC-333.
SC-D-1055	-	Handset Switch, Assembly and Details.
SC-B-14423	-	S-hook Assembly.
SC-D-22545	-	Handle, Handset Assembly.
SC-C-22546	-	Cap, Receiver (Plastic).
SC-C-22547	-	Cap, Transmitter (Plastic).
SC-B-22556	-	Transmitter.
SC-DL-34036	-	Connector Plug IJ-161( )/U.
SC-D-60464	-	Cord Assembly Electrical.
SC-DL-76644	-	Cord CD-494( ).
SC-C-101109	-	Cord Assembly Tinsel, Retractile, Group I
SC-C-105975	-	Switch Assembly.
SC-B-105976	-	Receiver.
SC-D-105992	-	Handset Assembly.

(Copies of the above drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D3955	-	Standard Specification for Electrical Insulating Varnishes

(Copies of these documents are available online at http://www.astm.org or from the ASTM International, P.O. Box C700, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 10012-1	-	Quality Assurance Requirements for Measuring Equipment, Metrological Confirmation System for Measuring Equipment.

(Copies of these documents are available online at http://www.iso.ch or from the International Organization for Standardization American National Standards Institute, 11 West 42nd Street, 13th Floor, New York, NY 10036.)

IPC - ASSOCIATION CONNECTING ELECTRONICS INDUSTRIES

J-STD-004	-	Requirements for Soldering Fluxes
J-STD-005	-	Requirements for Soldering Pastes-Replaces QQ-S-571
J-STD-006	-	Requirements for Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications.

(Copies of this document are available from http://www.ipc.org or IPC - Association Connecting Electronics Industries, 3000 Lakeside Drive, Suite 309 S, Bannockburn, IL 60015-1249.)

NATIONAL CONFERENCE OF STANDARDS LABORATORIES (NCSL)

NCSL Z540.1	-	Calibration Laboratories and Measuring and Test Equipment - General Requirements-Replaces Mil-Std-45662

(Copies of these documents are available from http://www.ncsli.org or to National Conference of Standards Laboratories (NCSL), 2995 Wilderness Place, Suite 107, Boulder, CO 80301-5404.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MW 1000	-	Magnet Wire

(Copies of these documents are available online at http://www.nema.org or from the National Electrical Manufacturer's Association, 1300 North 17th Street, Suite 1847, Rosslyn VA 22209.)

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheet. The individual Item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. Handsets furnished under this specification shall be products, which have been tested and have passed the first article inspection specified in 4.5.

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3.3 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the handsets to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Fungus resistant materials (see 4.7.11). The product shall be constructed of fungus-inert materials, and shall show no evidence of fungus or other corrosion, which may cause a mechanical failure. Following the above referenced test, the operational talk signal shall not degrade due to fungus and shall be in accordance with 3.8. The contractor may refer to requirement 4 of the military handbook titled “General Guidelines for Electronic Equipment”, for additional information and guidance on fungus-resistant materials (see 6.8).

3.3.2 Solder. There shall be no evidence of “cold soldering” and the use of excessive amounts of solder will not be acceptable. Solder shall be solid or rosin flux-cored in accordance with composition Sn97Pb3. Soldering materials and methods shall be as specified in J-STD-004, J-STD-005, and J-STD-006.

3.3.3 Pure tin. The use of pure tin, as an underplate or final finish, is prohibited both internally and externally. Tin content of handset components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass (see 6.5).

3.3.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Design and construction.

3.4.1 Wire, magnet. Magnet wire shall conform to NEMA MW 1000.

3.4.2 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings.

3.4.3 Connections. Soldering leads, lugs, and terminals shall be silver-plated or tin lead alloy coated. Wires subject to breakage at the connections shall be provided with terminals that grip the wire insulation. Solder-less terminals may be used on tin lead alloy coated standard wire, in power or audio circuits. But, other wire connections shall be soldered. Where practicable, wire soldered to terminals shall be looped at least once and not more than twice around the terminals before soldering, or equivalent means shall be employed. Textile insulation ends of wires shall be secured against fraying by mechanical means or by application of varnish conforming to ASTM D3955.

3.4.4 Earphone elements. The earphone elements shall be tested to and shall meet the first article and conformance requirements of MIL-M-13189.

3.4.5 Microphone elements. The microphone elements shall be tested to and shall meet the first article and conformance requirements of MIL-M-13189.

3.4.6 Switch assembly. The switch shall conform to the specified drawing (see 3.1).

3.4.7 Cord assembly. The cord and cord assembly shall conform to the specified drawing (see 3.1).

3.4.8 S-hook assembly (when applicable). The S-hook assembly shall conform to the specified drawing (see 3.1).

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3.4.9 Receiver. The receiver shall conform to the specified drawing (see 3.1).

3.4.10 Transmitter. The transmitter, transmitter cap, and receiver cap shall conform to the specified drawing (see 3.1).

3.4.11 Plugs. The plugs shall be as specified (see 3.1).

3.4.12 Handle. The handle shall conform to the specified drawing (see 3.1).

3.4.13 Screws. Screws shall be as specified (see 3.1).

3.4.14 Wire. The wire shall conform to the specified drawing (see 3.1).

3.4.15 Terminal. The terminal shall conform to the specified drawing (see 3.1).

3.5 Dielectric withstanding voltage. When tested as specified in 4.7.2, there shall be no evidence of degradation in performance of the handsets. There shall be no arcing or breakdown of the insulation.

3.6 Impedance (earphone). When tested as specified in 4.7.3, the impedance of the magnetic earphone shall be as specified (see 3.1).

3.7 Resistance (microphone)(see 4.7.4). When tested, the resistance shall be 28 $\pm$ 1.4 ohms.

3.8 Operational (talk). When handsets are tested as specified in 4.7.5, there shall be no buzzing, rattling, or other spurious sounds. There shall be transmission of an intelligible signal.

3.9 Vibration. When handsets are tested as specified in 4.7.6, there shall be no evidence of cracking, loosening of parts, or physical or electrical defects to the handset. Following the test, the earphones shall meet the frequency response requirements as specified in MIL-M-13189.

3.10 Moisture resistance. When handsets are tested as specified in 4.7.7, the handset shall operate in any relative humidity up to 98 percent without evidence of warpage or other damage. Corrosion, if present, shall not be sufficient to interfere with proper operation. Following the test, the earphones shall meet the frequency response requirements as specified in MIL-M-13189.

3.11 Random drop. When handsets are tested as specified in 4.7.8, there shall be no evidence of breaking, cracking, or other physical or electrical damage to the handset. Following the test, the earphones shall meet the frequency response requirements as specified in MIL-M-13189.

3.12 Temperature cycling. When handsets are tested as specified in 4.7.9, there shall be no evidence of warping or mechanical damage. Following the test, the earphones shall meet the requirements of frequency response as specified in MIL-M-13189.

3.13 Switches (when applicable). When handsets are tested as specified in 4.7.10, there shall be no evidence of physical damage, electrical defects or degradation in performance.

3.14 Marking. Handsets shall be marked in accordance with MIL-STD-130, with the type number and the manufacturer's name or symbol, and as specified (see 3.1).

3.15 Workmanship. Handsets shall be processed in such a manner as to be uniform in quality and shall be free from loose or deposited foreign materials, and other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Components inspection (see 4.3).
b. First article inspection (see 4.5).
c. Conformance inspection (see 4.6).

4.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with ISO-10012-1 and NCSL Z540.1. The inspection requirements specified herein are classified as follows.

a. Components inspection (see 4.3).
b. First article inspection (see 4.5).
c. Conformance inspection (see 4.6).

4.3 Components inspection. Components inspection shall consist of certification supported by verifying inspection and test data, when applicable, that the components listed in table I, which are used in fabricating the handsets, are in accordance with the applicable referenced document.

TABLE I. Components inspection.

Components	Requirement paragraphs	Applicable documents
Earphone elements Microphone elements Switch assembly	3.4.4 3.4.5 3.4.6	MIL-M-13189 MIL-M-13189 SC-D-1055 SC-C-105975 SC-D-105992
Cord assembly	3.4.7	SC-D-1041 SC-D-60464 SC-DL-76644 SC-C-101109
S-hook assembly	3.4.8	SC-B-14423
Receiver	3.4.9	SC-B-105976
Transmitter	3.4.10	SC-B-22556
Cap transmitter	3.4.10	SC-C-22547
Cap receiver	3.4.10	SC-C-22546
Plugs	3.4.11	MIL-P-642 SC-DL-34036
Handle	3.4.12	SC-D-22545 SC-D-105992

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4.4 Inspection conditions.

4.4.1 Conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" section of MIL-STD-202. All acoustical testing, except operational testing, shall be made in an anechoic chamber, which is the equivalent to the free field acoustic environment. Free field conditions are those that approach acoustically those of free space. An anechoic room should be considered satisfactory, if the sound pressure varies inversely as the distance, within ± 1 dB (decibel), over the measuring frequency range, under the following conditions: the sound source is effectively a point source (that is, emits spherical waves) and is located at the position or positions of the reference point[1/] of the device under test; the microphone is located at the position or positions it will occupy during the test; and the distance referred to is the distance between the reference point[1/] of the effective point source and the microphone.

4.5 First article inspection (see 3.2). First article inspection shall be performed by the supplier, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units, which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract or purchase order under which it is granted, unless extended by the Government to other contracts or purchase orders.

4.5.1 Sample size. Nine (9) handsets shall be subjected to first article inspection.

4.5.2 Inspection routine. The sample shall be subjected to the inspections specified in table II, in the order shown. All sample units shall be subjected to the inspections of group I. The sample shall then be divided equally into three groups and separately subjected to the inspections of groups II, III, and IV.

4.5.3 Failures. More than one failure in group I and any failure in group II, Ill, or IV shall be cause for refusal to grant first article approval.

1/ For single direct-radiator drivers having plane diaphragms, the reference point should be the center of the diaphragm; and for those having conical diaphragms, the reference point should be the center of a plane passing through the outer edge of the cone.

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TABLE II. First article Inspection.

Examination or test	Requirement paragraphs	Method paragraph
Group I		
Visual and mechanical	3.3, 3.4.1 thru 3.4.15 inclusive, 3.14 and 3.15	4.7.1
Dielectric withstanding voltage	3.5	4.7.2
Frequency response 1/		
Earphone	3.4.4	---
Microphone	3.4.5	---
Impedance (earphone)	3.6	4.7.3
Resistance (microphone)	3.7	4.7.4
Operational (talk)	3.8	4.7.5
Group II		
Vibration 1/	3.9	4.7.6
Moisture resistance 1/	3.10	4.7.7
Group III		
Random drop 1/	3.11	4.7.8
Temperature cycling 1/	3.12	4.7.9
Group IV		
Switch test	3.13	4.7.10

1/ Earphones and microphones shall meet the requirements of MIL-M-13189 after subjection to environmental tests.

4.6 Conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections. Except as specified in 4.6.1.4.4, delivery of products, which have passed groups A and B inspections, shall not be delayed pending the results of group C inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of all handsets of the same type, produced under essentially the same conditions, and offered for inspection at one time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examination and tests specified in table III, in the order shown.

4.6.1.2.1 Sampling plan. A sample of parts shall be randomly selected in accordance with table IV. If one or more defects are found, the lot shall be rescreened and defects removed.

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TABLE III. Group A inspection.

Examination or test	Requirement paragraph	Method paragraph
Visual and mechanical	3.3, 3.4.1 thru 3.4.15 inclusive, 3.14, and 3.15	4.7.1
Dielectric withstanding voltage	3.5	4.7.2
Operational (talk)	3.8	4.7.5

TABLE IV. Group A and group B sampling plans.

Lot size			Sample size	
			Group A	Group B
1	to	2	Entire lot	Entire lot
3	to	12	Entire lot	3
13	to	25	13	3
26	to	50	13	5
51	to	90	13	6
91	to	150	13	7
151	to	280	20	10
281	to	500	29	11
501	to	1,200	34	15
1,201	to	3,200	42	18
3,201	to	10,000	50	22
10,001	to	35,000	60	29
35,001	to	150,000	74	29
150,001	to	500,000	90	29
over 500,000			102	29

4.6.1.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for re-inspection. All failures occurring during testing shall be documented and investigated to determine root cause. Acceptance of the product shall be discontinued until corrective action, which is acceptable to the procuring activity, has been taken. After the corrective action has been taken, the group A inspections shall be repeated (all inspections, or the inspection which originally failed, at the option of the procuring activity). Approval shall be withheld until the inspection has shown that the corrective action was successful. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table V and shall be made on sample units, which have been subjected to and have passed group A inspection.

4.6.1.3.1 Sampling plan. A sample of parts shall be randomly selected in accordance with table IV. If one or more defects are found, the lot shall be rescreened and defects removed.

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TABLE V. Group B inspection.

Test	Requirement paragraph	Method, paragraph
Frequency response 1/		
Earphone	3.4.4	---
Microphone	3.4.5	---
Impedance (earphone)	3.6	4.7.3
Resistance (microphone)	3.7	4.7.4

1/ Handset components shall meet the requirements of MIL-M-13189.

4.6.1.3.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for re-inspection. All failures occurring during testing shall be documented and investigated to determine root cause. Acceptance of the product shall be discontinued until corrective action, which is acceptable to the procuring activity, has been taken. After the corrective action has been taken, the group B inspections shall be repeated (all inspections, or the inspection which originally failed, at the option of the procuring activity). Approval shall be withheld until the inspection has shown that the corrective action was successful. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.3.3 Disposition of sample units. Sample units which have passed the group B inspection may be delivered on the contract or purchase order, if the lot is accepted and the sample units are still within specified electrical tolerances.

4.6.1.4 Group C Inspection. Group C inspection shall consist of the tests specified in table VI, in the order shown. Group C inspection shall he made on sample units selected from inspection lots, which have passed groups A and B inspections.

4.6.1.4.1 Sampling plan. Group C inspection shall be performed once each month on 9 (nine) sample units selected without regard to their quality from units produced during the period or each 1,000 units, whichever occurs first. The sample shall be divided equally into three groups and subjected to the tests of subgroups 1, 2, and 3 of table VI.

TABLE VI. Group C inspection.

Test	Requirement paragraph	Method paragraph
Subgroup 1		
Vibration 1/	3.9	4.7.6
Moisture resistance 1/	3.10	4.7.7
Subgroup 2		
Random drop 1/	3.11	4.7.8
Temperature cycling 1/	3.12	4.7.9
Subgroup 3		
Switches	3.13	4.7.10

1/ After subjection to environmental tests, earphones and microphones shall meet the requirements of MIL-M-13189.

4.6.1.4.2 Failures. If one or more sample units fail to pass group C inspection, the sample shall be considered to have failed.

4.6.1.4.3 Disposition of sample units. Sample units, which have been subjected to group C Inspection, shall not be delivered on the contract or purchase order.

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4.6.1.4.4 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product, which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspections may be reinstituted. However, final acceptance shall be withheld until the group C re-inspection has shown that the corrective action was successful. In the event of failure after re-inspection, information concerning the failure and corrective action taken shall be furnished to the responsible inspection activity and the procuring activity.

4.7 Methods of examination and test.

4.7.1 Visual and mechanical examination. Handsets shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.3, 3.4.1 thru 3.4.15 inclusive, 3.14 and 3.15).

4.7.2 Dielectric withstanding voltage (see 3.5). The handset shall be tested with the microphone element and the earphone element removed. The handset shall be tested In accordance with method 301 of MIL-STD-202. The following details shall apply:

a. Test voltage - 440 volts.
b. Nature of potential - 60 Hz ac.
c. Points of application - The test voltage shall be applied between the wiring and adjacent metal nonconductors and ground of the wired handset handle.

4.7.3 Impedance (earphone) (see 3.6). The magnetic earphone elements shall be measured, while located on a 6 cm^3 coupler, as specified for the response test in MIL-M-13189. Connect the magnetic earphone element, in series with a non-inductive resistance at least 100 times the rated impedance of the magnetic earphone element, to a source of alternating voltage at 1,000 Hz. Adjust the source voltage to obtain an rms voltage across the magnetic earphone element, as specified in table VII as follows:

TABLE VII. Earphone Impedance test values.

Rated impedance of magnetic earphone elements	RMS voltage across magnetic earphone elements
ohms	volt
128	0.057
256	0.080
3,000	0.275

Replace the magnetic earphone element with a variable non-inductive resistance, and without changing the source voltage, adjust the variable resistance to obtain the same voltage as for the magnetic earphone element. The ohmic value of the resistance necessary to obtain the rms voltage reading shall be considered as the impedance of the magnetic earphone element.

4.7.4 Resistance (microphone, see 3.7). The resistance for the test shall be taken as the maximum resistance, (determined from the potential (voltage) dc and current dc readings), occurring during the periods from 5 seconds after the agitation is completed until the end of the test.

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4.7.5 Operational (talk) (see 3.8). Completely assembled handsets shall be tested by a two-way talk-test to insure correct wiring and satisfactory operation.

4.7.6 Vibration (see 3.9). The handset shall be tested for vibration in accordance with method 201 of MIL-STD-202. The following details shall apply:

a. Visual-mechanical examination shall be made prior to the vibration.
b. During the vibration, the handset shall be wired as specified in the operational (talk) test (see 4.7.5).
c. Method of mounting shall be as prescribed in the procedure of the test method.
d. After the test, the handset shall be examined for loose and defective parts and the frequency-response shall be measured, as specified in MIL-M-13189.

4.7.7 Moisture resistance (see 3.10). The complete handset shall be tested in accordance with method 106 of MIL-STD-202. The following details shall apply:

a. Mounting - Any convenient mounting with the front face of microphone cap and the receiver cap exposed, and facing the air current, if any.
b. Polarizing voltage - Not applicable.
c. Step 7b - Not applicable.
d. Final measurements - At the completion of the tenth cycle and following a 24-hour period at 25 ± 5 degrees C and 50 ± 5 percent relative humidity, the handset shall be tested for frequency response in accordance with MIL-M-13189 and examined for loose or deformed parts, corrosion, or other damage.

4.7.8 Random drop (see 3.11). The handset shall be dropped six times from a height of 4 feet onto a concrete floor. The handset shall strike at least once on the receiver and once on the transmitter end. The frequency response shall be measured, as specified in MIL-M -13189, and the headset shall be examined for damage.

4.7.9 Temperature cycling (see 3.12). The equipment shall be tested in accordance with method 102 of MIL-STD-202. The following details shall apply:

a. Test condition letter - D.
b. Measurement after cycling - The equipment shall be examined for damage and the frequency response shall be measured, as specified in MIL-M-13189.

4.7.10 Switch test (when applicable, see 3.13). The switch shall be subjected to 100,000 make-and-break cycles at a rate not to exceed 12 cycles per minute. Tests requiring repetitive operation of switches shall be made using a suitable device arranged to move the switch lever into each “ON” position and back to the original position to complete one cycle of operation, and thereby causing the switching mechanism to make-and-break (or break-and-make) all the circuits, in which the switch is connected.

4.7.11 Fungus (see 3.3.1). At the option of the contractor, the contractor shall certify that the materials are fungus resistant materials, or test method 508 of MIL-STD-810 shall be performed.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The handsets covered by this specification are intended to be used with wire and radio communication equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of the applicable specification sheet and the applicable type number (see 1.2.1 and 3.1).
c. Packaging (see 5.1).
d. First article inspection, when required (see 3.2).

6.3 First article inspection. Information pertaining to first article inspection of components covered by this specification should be obtained from the procuring activity for the specific contracts involved (see 3.2).

6.4. Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/PIQ-1W.

6.5 Tin whisker growth (see 3.3.3). The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM B545 (Standard Specification for Electrodeposited Coatings of Tin).

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6.6 Subject term (key word) listing.

Acoustical
Anechoic chamber
Cord
Decibel
Earphone
Frequency response
Handle
Magnet wire
Microphone
Radio communication
Receiver
Switches
Transmitter

6.7 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. Table VIII lists the Environmental Protection Agency (EPA) top seventeen hazardous materials targeted for major usage reduction. Use of these materials should be minimized or eliminated, unless needed to meet the requirements specified herein (see section 3).

TABLE VIII. EPA top seventeen hazardous materials.

Benzene	Dichloromethane	Tetrachloroethylene
Cadmium and Compounds	Lead and Compounds	Toluene
Carbon Tetrachloride	Mercury and Compounds	1,1,1 - Trichoroethane
Chloroform	Methyl Ethyl Ketone	Trichloroethylene
Chromium and Compounds	Methyl Isobutyl Ketone	Xylenes
Cyanide and Compounds	Nickel and Compounds	

6.8 Fungus resistant materials (see 3.3.1). The contractor may refer to requirement 4 of the military handbook MIL-HDBK-454 titled “General Guidelines for Electronic Equipment”, for additional information and guidance on fungus-resistant materials.

6.9 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA – CC

Preparing activity
DLA – CC

(Project 5965-0372-000)

Review activities:
Army – SM
Navy – CG, MC, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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